

A Human-Centered Workflow for Using Large Language Models in Content Analysis

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Generative Artificial Intelligence in Qualitative Data Analysis: Analyzing—Or Just Chatting?

Organizational Research Methods

1–37

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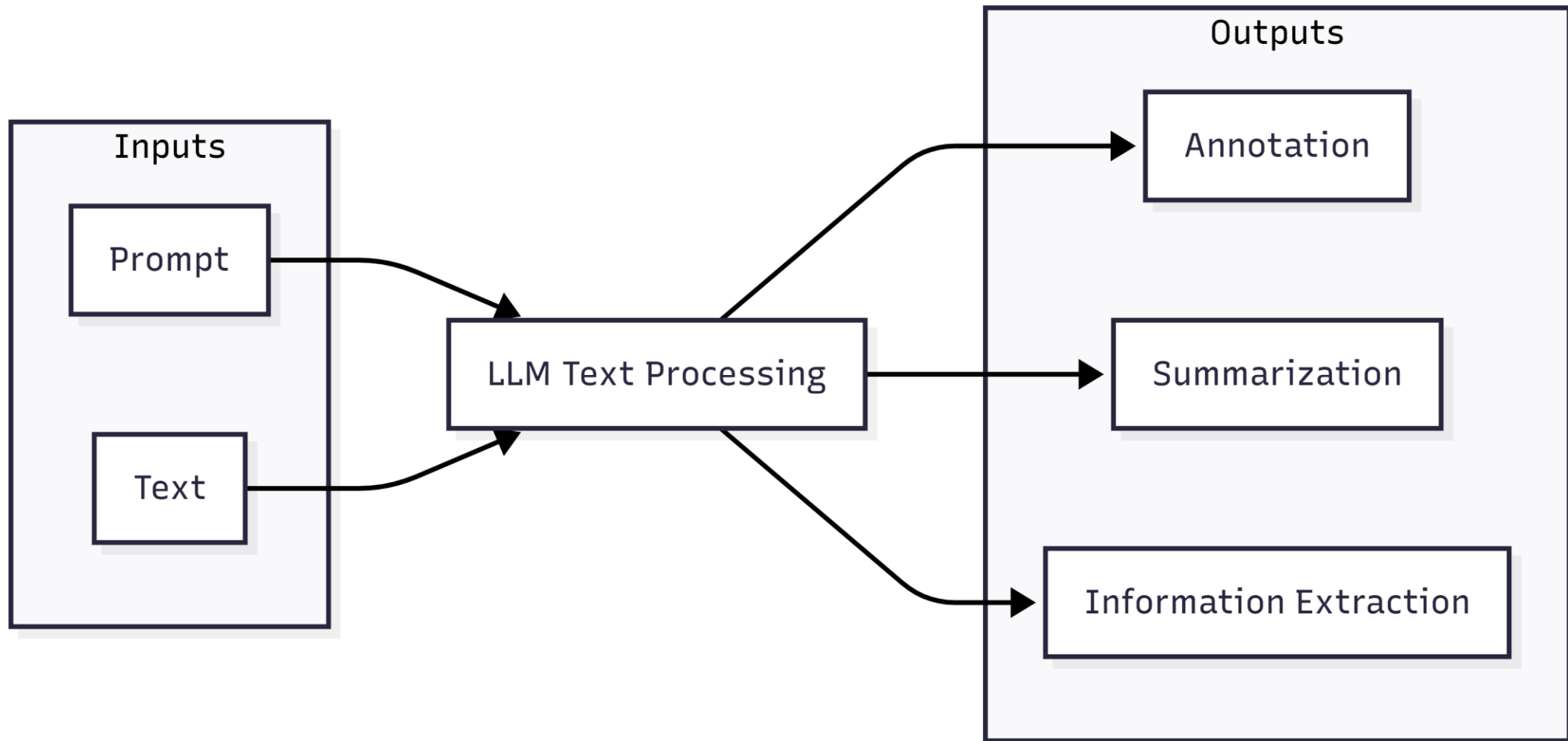
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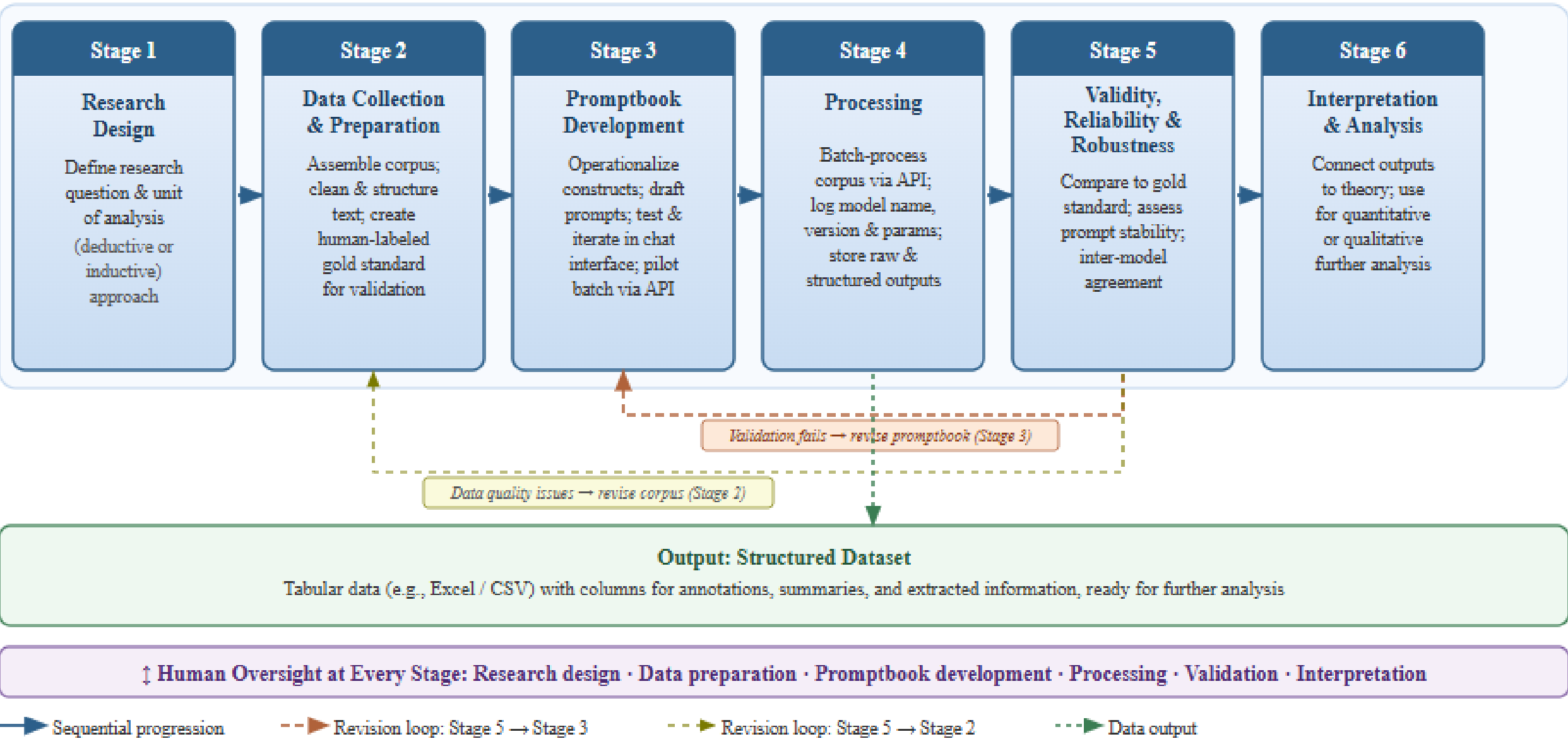
Conclusion: “**GenAI**—based on the current transformer architecture on which all foundational LLMs are built—is **unsuited to qualitative data analysis** and presents multiple epistemic risks to management research were its adoption sanctioned.”

Introduction

- Huge interest in using LLMs in content analysis workflows (Barros et al., 2025; Than et al., 2025; Törnberg, 2024)
 - Mostly in Politics and Sociology
- A **human-centered workflow**
- The real power of LLMs is in using them through **API**
- Contributions:
 - Development of a **rigorous & repeatable process**
 - Synthesis of **vast methodological literature** on LLMs for Qualitative research (mostly from Politics & Sociology)
 - **Practical resources for researchers:** Python Code & prompt library to implement the process

Large Language Models as Universal Text Processing Machines, programmable by Prompt





1. Research design

- Identify which parts of the analysis could be augmented by the LLM
- Consider any ethical implications and biases
- LLMs can help to brainstorm
- What do you want from an LLM: efficiency, scale, breadth, consistency, innovativeness, triangulation (checking if an AI finds the same patterns you do)

2. Data collection & preparation

- **Data sources:** Interviews / Social media / Web pages / Company archives / Databases
- **Data collection methods:** primary collection / API / web scraping / databases / ...
- Make sure your data is representative of the population you study
- Convert to either files (e.g., pdf) or tabularic format (e.g., Excel)
- Transcription & cleaning
- Anonymization
- Segmenting data if too large
- Pre-analysis by human researcher

3. Promptbook development

- Prompt engineering & codebook definition => Promptbook
- Prompting types: **zero-shot**, few shot, **persona**, chain-of-thought, chain-of-density...
- **Iterative process**
- Have some smaller subset of data where you know the answer
- Use **different models** through chat
- When it works in chat, **test through API** on smaller subset
- If it doesn't work, change the prompt & test again

Well-structured prompts

The essential components of a well-structured prompt include:

- **Role Specification.** Instructing the model to adopt a specific persona (e.g., "You are a management academic" or "You are an expert in Leadership") can prime it for the task (Pavlovic & Poesio, 2024; Than et al., 2025).
- **Task Directive.** A clear, unambiguous command that outlines the core objective of the annotation.
- **Context and Definitions.** This section operationalizes the concepts of interest by providing detailed definitions, inclusion/exclusion criteria, and examples for handling ambiguous cases, drawn directly from the research codebook (Dunivin, 2025; Halterman & Keith, 2025).

Well-structured prompts

The essential components of a well-structured prompt include:

- **Response Formatting.** Specifying the desired output structure (e.g., a single label, a number, table row, or a JSON object) is crucial for ensuring consistent results that can be parsed and stored in a database, typically in a tabular form (e.g., Excel sheet) (Carlson & Burbano, 2025; Stuhler et al., 2025).
- **Built-in triangulation.** Good prompt design involves building the prompt in a way that enables initial triangulation within the results before checking the original documents. LLMs can be asked to explain their decisions and copy verbatim text from documents relevant for specific code/annotation. In authors' experience models rarely hallucinate when instructed to copy verbatim text from the document.
- **Missing information instructions.** Explicitly instruct the model what to do if the requested information is not found in the text (e.g. "Return N/A if none.").

You are a researcher in Management conducting a literature review. You will receive a text of an academic paper. Please use the following coding manual to code the text.

- IE_TITLE = Title of the paper.
- AN_RQ = Did the authors explicitly define research questions, research aims, or research objectives? (1 = Yes, 0 = No).
- SU_RQ_VERBATIM = Copy verbatim text from the paper that defines the research questions/aims/objectives. If none, return N/A.
- SU_FULL_PROCESS = What were the steps of the research and analysis process from start to finish?
- SU_KEY_CONTRIBUTIONS = What are the key theoretical contributions of the paper?
- AN_PAPER_TYPE = Type of the paper – empirical, review, conceptual, editorial, other.
- SU_PAPER_TYPE_REASON = Explain why you chose the paper type.
- IE_SAMPLE_SIZE = What was the size of the sample? Return N/A if not explicitly stated.

Put all extracted variables in a downloadable Excel table

Developing an effective promptbook

- **Step 1: Draft the output structure.** Begin by designing a draft output table based on the study's research aims and questions. Each column should correspond to a distinct analytical dimension that the model will be asked to address.
- **Step 2: Develop the promptbook.** Write a draft promptbook in which each column has a clearly specified task instruction and an associated variable name. Prompts should be unambiguous, precise, and constructive, guiding the model toward desired behaviors rather than specifying what to avoid (Khalid & Witmer, 2025).

Developing an effective promptbook

Step 3: Iterative testing in a chat interface.

- Test different versions of the promptbook in a chat interface across a small number of documents that ideally represent the diversity of the full corpus.
- Test across multiple models (both within and across providers) is advisable, as performance can vary considerably (Boonstra, 2025).
- Context window management:
 - When applying the prompt to a new document, the context window must be reset, since prior text may otherwise contaminate the model's response.
 - The simplest way to do this is to begin a new chat session for each document.
- LLMs themselves can help with promptbook development. It is often useful to ask LLM to improve the prompt and/or suggest additional variables to annotate/summarize/extract.
- If output quality is unsatisfactory, return to Steps 1 and 2 and revise accordingly. This cycle should be repeated until the model's responses are consistently adequate.

Developing an effective promptbook

Step 4: Pilot API batch processing.

- Once the prompts perform well in the chat interface, proceed to test batch processing via the API on a small random sample of documents (e.g., 10–20).
- Formally validate the batch output using appropriate reliability and validity metrics (see the section on assessing validity, reliability, and robustness).
- If performance falls short, return to Steps 1–3 with particular attention to the tasks and document types that failed validation. Repeat until validation metrics reach satisfactory levels.

4. Processing

- Choosing a platform (Google, Open AI, ...) vs running on local machine
- Batching and rate limits
- Error handling
- Audit trail (which prompt was used and when, which code...)

5. Assessing validity, reliability and robustness

- Ground truth (human raters)
- Precision, Recall, F1
- Inter-coder/Inter-model agreement (Human vs AI; AI model 1 vs AI model 2)
- Multiple prompt variations? (prompt robustness)
- Check for hallucinations (errors)
- Humans can interpret

Metric	Definition	Use Case	Validity/Reliability/Robustness
Accuracy	Proportion of correct predictions	Overall performance on balanced datasets	Validity
Precision	True positives / (True positives + False positives)	When false positives are costly, we want to maximize precision.	Validity
Recall	True positives / (True positives + False negatives)	When false negatives are costly, we want to maximize recall.	Validity
F1 Score	Harmonic mean of precision and recall	Balanced measure for imbalanced classes	Validity
Mean Absolute Error (MAE)	Average absolute difference from true values	Continuous outcome validation	Validity
Cohen's Kappa	Agreement beyond chance between raters	Inter-rater or inter-model agreement	Reliability
Krippendorff's Alpha	Reliability measure handling multiple raters/missing data	Complex annotation schemes	Reliability
Intraclass Correlation (ICC)	Consistency of continuous ratings across runs	Repeated measurements	Reliability
Prompt Stability Score	Consistency across semantically similar prompts	Assessing prompt robustness	Robustness

6. Interpretation & Further analysis

- Interpretation reconnects model outputs to theory and to the underlying texts.
- For **quantitative content analysis**, this involves treating LLM outputs as variables and evaluating whether inferences are robust across plausible prompts/models and consistent with validation evidence.
- For **qualitative analysis**, interpretation requires returning to the text: using LLM outputs to navigate, retrieve, and compare passages, while maintaining reflexive memoing and documenting how human judgments shaped category refinement.
- In both cases, the standard should be **traceability**: readers should be able to see how constructs were operationalized, how evidence was extracted, and where human interpretation entered the chain.

Discussion & Conclusion

- Concerns:
 - Accuracy & reliability
 - Reproducibility
- Conclusion:
 - LLMs can be **powerful assistants or co-analysts, but not replacements for human researchers**. Only humans can interpret and validate the findings.
 - The ideal approach is a **Human-AI partnership**

Thank you!

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